Business Performance & Operational Analytics Dashboard

Executive Summary

This project develops a reproducible Python workflow to clean operational order records, calculate business KPIs, compare segments, and visualize revenue, profit, delivery performance and customer satisfaction.

1. Problem Statement

Business performance information is often distributed across categories, regions, and departments. This project consolidates those dimensions to support monitoring, performance comparisons, and operational investigation.

2. Dataset and Scope

The synthetic dataset contains 1,501 raw order records (12 original fields) and 1,497 cleaned records (15 fields including derived measures). Coverage: 2025-01-01 to 2026-08-31. Fields include order date, region, department, product category, revenue, cost, returns, delivery days and satisfaction.

3. Tools and Technologies

Python 3, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook / Google Colab.

4. Methodology

Data loading and inspection; duplicate removal and quality checks; missing-value treatment; calculation of profit, profit margin, and order month; KPI calculations; monthly/category/region/department aggregations; exploratory visualization; export of reusable CSV outputs.

5. Key Performance Indicators

Metric	Value
Total Revenue	₹17,652,945.51
Total Cost	₹11,548,792.11
Total Profit	₹6,104,153.40
Profit Margin	34.58%
Unique Orders	1,497
Average Order Value	₹11,792.21
Return Rate	6.55%
Average Delivery Time	3.81 days
Average Satisfaction	4.13/5

6. Dashboard and Visual Outputs

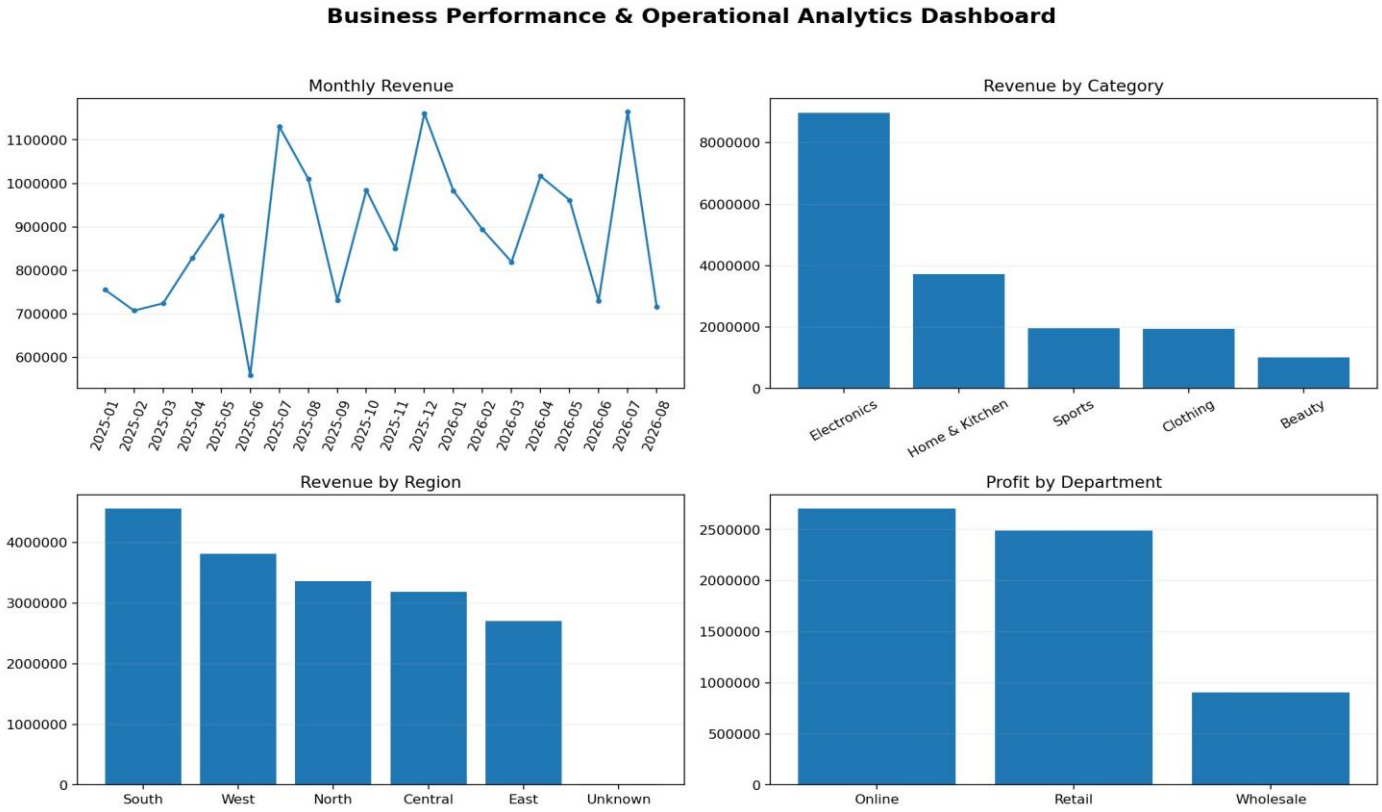


Figure 1. Monthly revenue, revenue by category, revenue by region, and profit by department (generated from exported summaries).

7. Findings

Electronics is the largest revenue category (₹8,982,933.12). South leads regions by revenue (₹4,558,331.02). Beauty has the lowest product-category profit margin (34.15%). Clothing has the highest observed category return rate (8.61%).

The Unknown region has only two orders; its average delivery time should not be interpreted as a stable regional benchmark.

8. Recommendations

Review pricing and cost structure in low-margin categories. Examine the categories with elevated return rates. Investigate regional delivery differences with appropriate sample-size checks. Monitor monthly KPI changes against documented business targets.

9. Results and Conclusion

The project delivered cleaned order data, nine headline KPIs, multiple performance summaries, and an integrated four-panel dashboard. These outputs support descriptive comparison and provide a foundation for future business reporting.

10. Limitations

All observations are based on synthetic educational data. The analysis is descriptive and does not establish causes or claim improvements in revenue or efficiency. Missing marketing, staffing, and inventory details limit operational diagnosis.